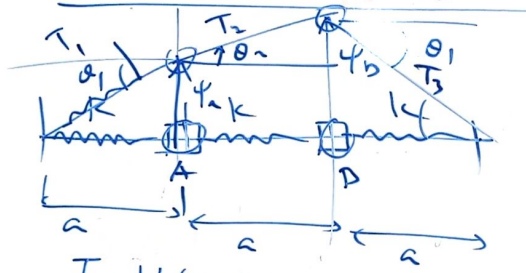
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Transverse osc. in two coupled mass system



$T_0 = K(a - a_0)$

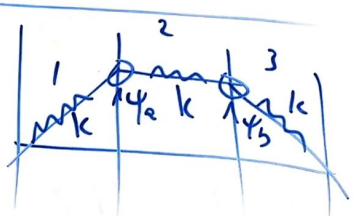
For small osc.,

$T_1 \approx T_2 \approx T_3 \approx T_0$

$$M\ddot{\psi}_a = T_2 \sin\theta_2 - T_1 \sin\theta_1$$
$$\approx T_0 \tan\theta_2 - T_0 \tan\theta_1$$
$$= T_0 \frac{\psi_b - \psi_a}{a} - T_0 \frac{\psi_a}{a}$$

$$M\ddot{\psi}_b = -T_3 \sin\theta_3 - T_2 \sin\theta_2$$
$$\approx -T_0 \tan\theta_3 - T_0 \tan\theta_2$$
$$= -T_0 \frac{\psi_b}{a} - T_0 \left(\frac{\psi_b - \psi_a}{a}\right)$$

Mode 1 :

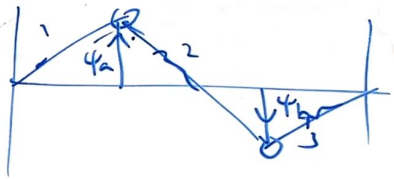


$\psi_b = \psi_a \Rightarrow \ddot{\psi}_a = -\frac{T_0}{Ma}\psi_a = -\omega_1^2 \psi_a$

Restoring force comes from spring 1 and 3

$\omega_1^2 = \frac{T_0}{Ma}$

Mode 2 : $\psi_b = -\psi_a$



$$\ddot{\psi}_a = -\frac{3T_0}{Ma}\psi_a = -\omega_2^2 \psi_a$$

$$\omega_2 = \sqrt{\frac{3T_0}{Ma}}$$

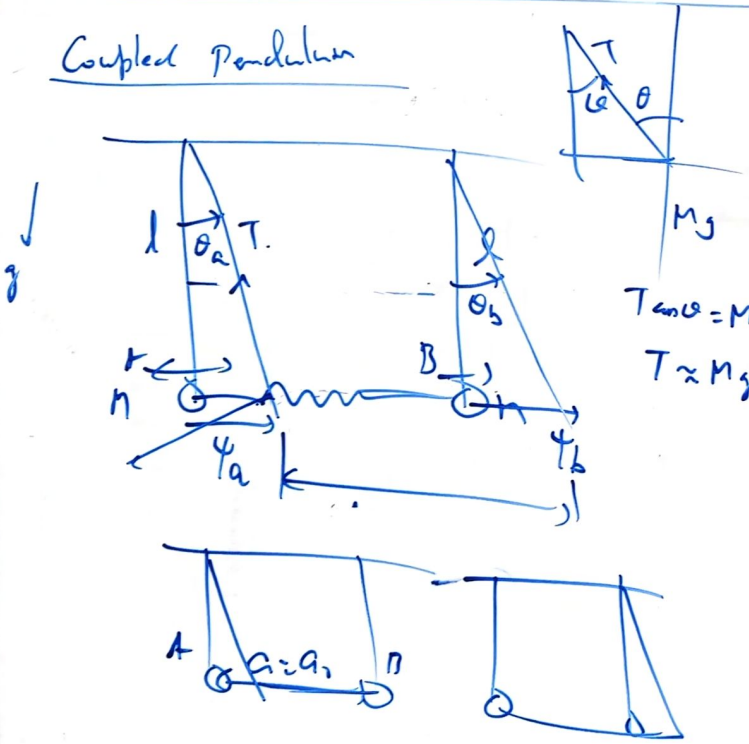
Mode 1 : $\frac{\psi_b}{\psi_a} = 1 \Rightarrow \frac{B_1}{A_1} = 1$ or $B_1 = A_1$

Mode 2 : $\frac{\psi_b}{\psi_a} = -1 \Rightarrow B_2 = -A_2$

$\psi_a = A_1 \cos(\omega_1 t + \varphi_1) + A_2 \cos(\omega_2 t + \varphi_2)$

$\psi_b = A_1 \cos(\omega_1 t + \varphi_1) - A_2 \cos(\omega_2 t + \varphi_2)$

Coupled Pendulum



$T\cos\theta = Mg$

$T \approx Mg$

$m\ddot{\psi}_a \approx -T\tan\theta_a + K(\psi_b - \psi_a)$

$\approx -Mg\frac{\psi_a}{l}$

(1) $\ddot{\psi}_a \approx -\frac{g}{l}\psi_a + \frac{K}{m}(\psi_b - \psi_a)$

$\ddot{\psi}_b \approx -\frac{g}{l}\psi_b + \frac{K}{M}(\psi_a - \psi_b)$

(2) $\ddot{\psi}_b = -\frac{g}{l}\psi_b - \frac{K}{M}(\psi_b - \psi_a)$

Mode 1: $\psi_b = \psi_a$,

$\omega_1^2 = g/l$

Mode 2: $\psi_b = -\psi_a$;

$\ddot{\psi}_a = -\underbrace{\left(\frac{g}{l} + \frac{2K}{m}\right)}_{\omega_2^2}\psi_a$

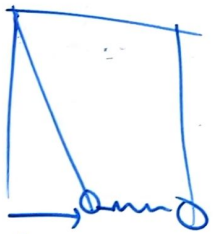
$\omega_2^2 = g/l + 2K/m$

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Coupled Pendulum

(1) $\psi_a = A_1 \cos(\omega_1 t + \varphi_1) + A_2 \cos(\omega_2 t + \varphi_2)$

(2) $\psi_b = A_1 \cos(\omega_1 t + \varphi_1) - A_2 \cos(\omega_2 t + \varphi_2)$



$\psi_a(0) = 2A$

ICs: $\psi_a(0) = 2A, \quad \psi_b(0) = 0$

$\dot{\psi}_a(0) = 0, \quad \dot{\psi}_b(0) = 0$

(3) $\psi_a(0) = A_1 \cos\varphi_1 + A_2 \cos\varphi_2 = 2A$

(4) $\psi_b(0) = A_1 \cos\varphi_1 - A_2 \cos\varphi_2 = 0$

(3), (4) $\Rightarrow$ $A_1 \cos\varphi_1 = A, \quad A_2 \cos\varphi_2 = A$ — (5)

(6) $\dot{\psi}_a(t) = -\omega_1 A_1 \sin(\omega_1 t + \varphi_1) - \omega_2 A_2 \sin(\omega_2 t + \varphi_2)$

(7) $\dot{\psi}_b(t) = -\omega_1 A_1 \sin(\omega_1 t + \varphi_1) + \omega_2 A_2 \sin(\omega_2 t + \varphi_2)$

(8) $\dot{\psi}_a(0) = -\omega_1 A_1 \sin\varphi_1 - \omega_2 A_2 \sin\varphi_2 = 0$

(9) $\dot{\psi}_b(0) = -\omega_1 A_1 \sin\varphi_1 + \omega_2 A_2 \sin\varphi_2 = 0$

(8), (9) $\Rightarrow$ $\omega_1 A_1 \sin\varphi_1 = 0, \quad \omega_2 A_2 \sin\varphi_2 = 0$

$\Rightarrow$ $\varphi_1 = 0, \quad \varphi_2 = 0$ (10)

$\Rightarrow$ $\cos\varphi_1 = 1, \quad \cos\varphi_2 = 1$

$\Rightarrow$ $\boxed{A_1 = A_2 = A}$